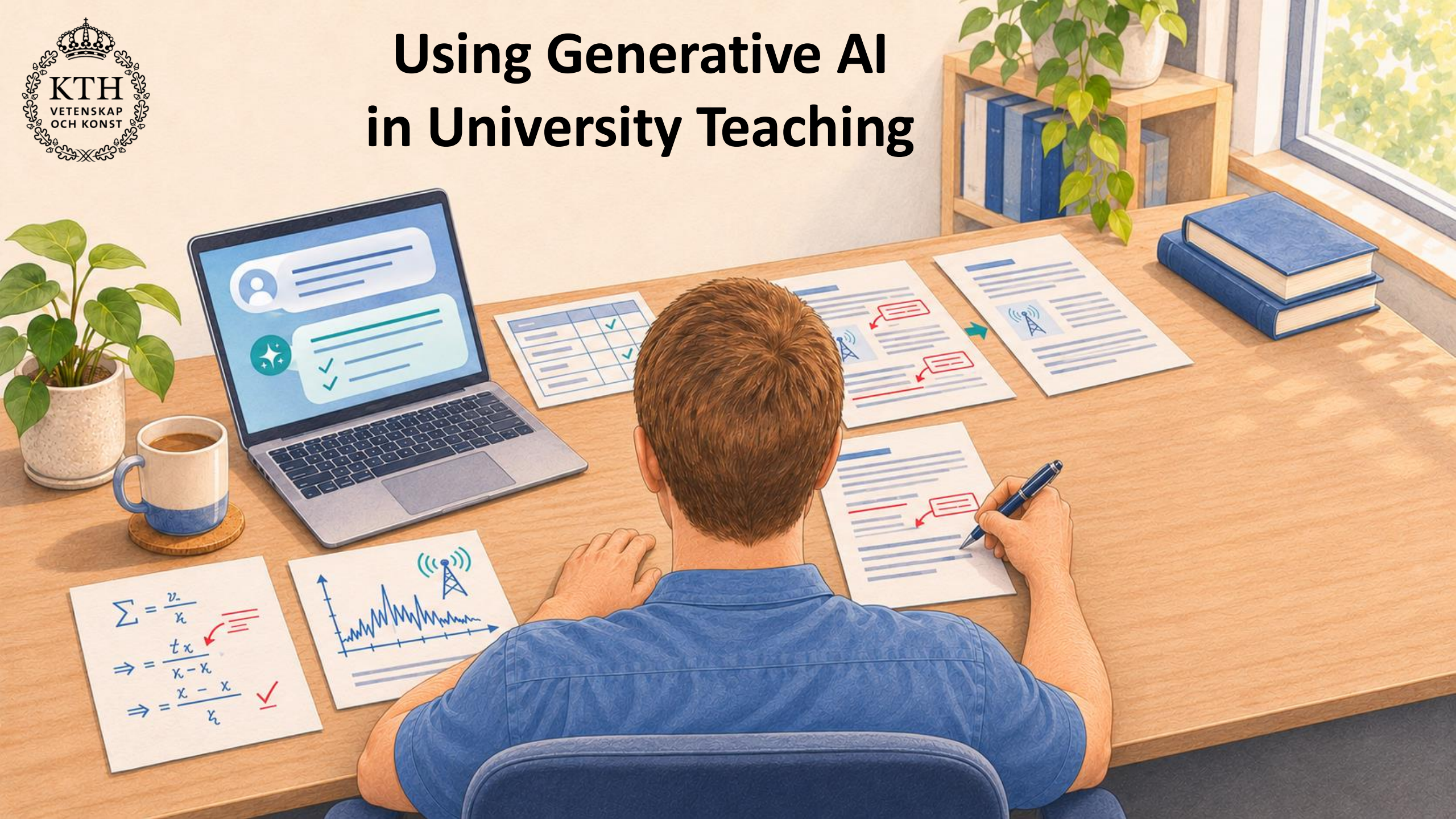
Using Generative AI in University Teaching



MY MAIN GOALS

1. Improve teaching and assessment quality
2. Save time on repetitive tasks, giving room for more expert work
3. Strengthen student learning and critical use of AI

USE OF GENERATIVE AI	EXAMPLE	INTENDED BENEFIT
Grading rubrics	Align with learning outcomes. Calibrate against earlier reports and grades. Draft guidance for students and teachers.	Quality and fairness More consistent grading
Student templates	Structure reports and proposals around research questions, methods, assumptions, and limitations.	Clarity and efficiency Fewer misunderstandings
Assessment design	Adapt tasks for AI use. Include staged submissions and individual explanations of modelling choices.	Assessment validity Clearer evidence of understanding

COURSE PREPARATION

USE OF GENERATIVE AI	EXAMPLE	INTENDED BENEFIT
Review teaching material	Check slides for typos, inconsistent notation, missing assumptions, and errors in equations or units.	Quality and efficiency Clearer teaching material
Review student reports	Flag gaps in logic and unsupported claims. The teacher verifies technical details and makes the final judgment.	Quality and efficiency More focused teacher feedback
Exercise: Simulated interview	Students prepare questions, then interview an LLM playing a specified type of person.	Practice and AI literacy Rehearse interviewing
Exercise: Peer review and revision	Students write a report, obtain LLM peer review, then revise it before receiving teacher feedback.	Learning quality Practise critical revision

TEACHING DURING THE COURSE

USE OF GENERATIVE AI	EXAMPLE	INTENDED BENEFIT
Multiple-choice questions	Inspired by earlier exams and learning outcomes. Build distractors around common misconceptions.	Efficiency and quality Broader question bank
Mathematical problems	Draft problems and worked solutions. Verify derivations and numerical results.	Efficiency and accuracy More reusable problems
Exam quality review	Check topic coverage, difficulty, timing, ambiguous wording, and whether questions have unique answers.	Quality and fairness Fewer examination errors
Marking schemes	Draft partial-credit guidance and acceptable alternative solutions. Check consistency with the rubric.	Efficiency and fairness More consistent marking

WRITTEN EXAMINATION

Do you have further ideas?

Share in the comments!

